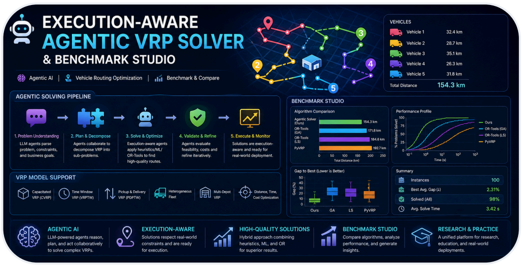
Industrial Project 2026-04-23

Execution-Aware Agentic VRP Solver and Benchmark Studio



Production-grade Vehicle Routing Problem solver with 8 solver backends, 18 operational scenarios, and an Agentic AI layer powered by Google Gemini 2.0 Flash.

Project Snapshot

DATE

2026-04-23

CATEGORY

Industrial Project

SHORT DESCRIPTION

Production-grade Vehicle Routing Problem solver with 8 solver backends, 18 operational scenarios, and an Agentic AI layer powered by Google Gemini 2.0 Flash.

TAGS / TECH STACK

AI · VRP · Vehicle Routing · OR-Tools · PyVRP · Agentic AI · Gemini · Streamlit · Optimization · MLOps · Hugging Face

Project Links

Demo Video · Hugging Face Space · GitHub Link · Project Link · GitHub Link · GitHub Link · Demo Video

Full Project Content

> Building an AI system that doesn't just **solve routes** — but combines 8 solver backends, 18 real-world scenarios, an Agentic AI orchestrator, and a benchmark-grade Streamlit studio into a single execution-aware optimization platform.

GitHub Repository: [dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio]

> 🎥 **Live demo video:** youtu.be/FqVuVjW20yo
>
> 🚀 **Try it live:** Hugging Face Space

Vision

Most VRP solvers stop at producing a route plan.

This project focuses on the full operational chain required to make routing decisions usable in a **live dispatch environment** where technicians are already in motion, jobs get cancelled mid-day, urgent orders inject themselves into the schedule, and shifts change without notice:

- Ingest 40,000 orders, 400 routes, 8 depots, multi-customer goal profiles
- Run 8 different solver backends in parallel (greedy, metaheuristic, constraint programming, cloud APIs)
- Stress-test against 18 operational scenarios (cancellations, sick technicians, urgent injects, SLA shifts)
- Lock in-progress routes so re-optimization preserves real-world execution state
- Expose an Agentic AI layer that detects violations and triggers re-optimization autonomously
- Provide a benchmark studio with KPI charts, interactive maps, and Excel export
- Deploy to Hugging Face Spaces with pre-computed artifacts for instant demo

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Project Attributes

Attribute	Description
`problem-statement`	Standard VRP solvers re-plan from scratch on every disruption — they're not aware that some stops are *already in execution*, leading to wasteful re-routing of stable assignments.
`primary-objective`	Build an execution-aware VRP platform that compares 8 solver families, locks in-progress work, and lets an AI agent autonomously fix constraint violations.

Attribute	Description
`core-technologies`	Python 3.11, Streamlit, Google OR-Tools, PyVRP, OSRM, Esri ArcGIS, Google Gemini 2.0 Flash, Folium, Plotly, Docker, Hugging Face Spaces.
`repository-scope`	8,140 lines across 4 source files · 18 fixed test scenarios · benchmark dataset · pre-computed artifacts · architecture docs · full function reference.
`runtime-interface`	Streamlit 6-tab UI: Overview · KPI Comparison · Map Explorer · Scenario Tables · Excel Export · 🤖 AI Agent.
`deployment-target`	Hugging Face Spaces (Docker SDK) for public demo · Self-hosted Streamlit for production.
`demo-surface`	Public HF Space with auto-loaded synthetic dataset and pre-computed benchmark artifacts.
`key-capabilities`	8 solver backends, locked-prefix re-optimization, agentic AI with 8 tools, interactive route maps, multi-sheet Excel export, reactive event handling.
`security-controls`	Environment-driven API keys (GOOGLE_API_KEY) · Optional Esri portal authentication · No hardcoded credentials.
`production-focus`	Reproducible benchmarks · 100% assignment guarantees · zero-violation winner backend · sub-1000s runtime · cloud-deployable Docker image.

Project Structure

```
src/content/posts/
└── execution-aware-agentic-vrp/
    ├── cover.png
    └── index.mdx

GitHub Repo: dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio/
├── app.py                           # Streamlit UI · 6 tabs · 845 lines
├── base_engine.py                   # 5 agents + scenario engine · 933 lines
├── compare_backends.py              # 8 solver backends + helpers · 5,438 lines
├── vrp_agent.py                     # Gemini agent + 8 tools · 960 lines
├── Dockerfile                       # HF Spaces deployment
├── requirements.txt
├── data/                            # Auto-loaded on startup
│   ├── orders_40000.csv
│   ├── routes_400.csv
│   ├── depots_8.csv
│   ├── accounts.csv
│   └── goal_profiles.csv
```

```
├── artifacts/                      # Pre-computed benchmark results (8 backends)
│   ├── OR-Tools_Execution-Aware/
│   │   └── vrp_scenario_results_full.xlsx
│   ├── OR-Tools/
│   ├── Hybrid_Execution-Aware_Rolling_VRP_Solver/
│   ├── Adaptive_Execution-Aware_Metaheuristic_Solver/
│   ├── Impacted-Subset_Greedy_2-opt/
│   ├── PyVRP/
│   ├── OSRM/
│   └── Esri/
└── docs/
    ├── ARCHITECTURE.md
    ├── ALGORITHM_COMPARISON.md
    └── REFERENCE_*.md (4 function reference files)
```

Why This Matters

> **Note** A vehicle routing solver becomes operationally useful only when it is **execution-aware, comparable, and self-correcting**.

> **Important** Real dispatch environments are not static — orders cancel, technicians call in sick, urgents inject mid-day. This project covers the missing layer between **batch route planning** and **live re-optimization with locked execution state**.

> **Tip** The 8-backend benchmark format generalizes to any combinatorial routing problem — replace the scenarios and orders, the comparative scaffold remains useful.

> **Warning** Lowest-distance solutions are not always best — PyVRP achieves the shortest routes (114 km) but produces 134,483 minutes of lateness. Constraint adherence matters as much as distance.

> **Caution** Public demo deployments should never use production API keys, real customer data, or unrestricted solver access. Use the included synthetic dataset for showcases.

What This System Does

```
flowchart TD
    A[5 CSV files: orders, routes, depots, accounts, goal_profiles] -->
B[load_inputs_from_uploads]
    B --> C[build_impacted_subset · 150 orders · 20 routes]
    C --> D[ScenarioAgent applies 1 of 18 mutations]

    D --> E{Backend Solver Selection}
    E --> F1[OR-Tools EA · WINNER]
    E --> F2[OR-Tools]
    E --> F3[Hybrid]
    E --> F4[Adaptive]
    E --> F5[Greedy + 2-opt]
    E --> F6[PyVRP]
    E --> F7[OSRM]
    E --> F8[Esri ArcGIS]

    F1 & F2 & F3 & F4 & F5 & F6 & F7 & F8 --> G[_build_common_bundle]
    G --> H[Output Bundle: 8 DataFrames]

    H --> I[Streamlit UI · 6 tabs]
    H --> J[Excel Export · 32 sheets]
    H --> K[Interactive Folium Maps]
    H --> L[VRPOrchestrator · Gemini Agent]

    L --> M[8 Tools · Autonomous Re-optimization]
    M --> H
```

Agentic AI Request Flow

```
sequenceDiagram
    autonumber
    participant U as User / Dispatcher
    participant ST as Streamlit UI
    participant ORC as VRPOrchestrator
    participant G as Gemini 2.0 Flash
    participant T as Tool (Python fn)
    participant CB as compare_backends.py

    U->>ST: "A driver called in sick — fix it"
    ST->>ORC: handle_event("RT_6", reason)
    ORC->>G: send_message + 8 tool definitions

    loop Up to 10 iterations
        G-->>ORC: FunctionCall(detect_violations)
        ORC->>T: dispatch_tool
        T-->>ORC: Late routes + amounts
```

```
        ORC->>G: FunctionResponse

        G-->>ORC: FunctionCall(suggest_config)
        ORC->>T: dispatch_tool
        T-->>ORC: Config overrides + reasoning
        ORC->>G: FunctionResponse

        G-->>ORC: FunctionCall(trigger_reoptimization)
        ORC->>CB: run_backend_suite(OR-Tools EA, new_config)
        CB-->>ORC: New bundle
        ORC->>G: Before/after diff
    end

    G-->>ORC: Final text summary
    ORC-->>ST: "Re-routed 8 orders to neighbouring techs. Distance +3.2 km, lateness 0."
    ST-->>U: Display in chat
```

Core Capabilities

- 8 solver backends spanning greedy, metaheuristic, constraint programming, and cloud APIs
- 18 fixed operational scenarios for reproducible benchmarking
- Locked-prefix re-optimization that preserves in-progress technician work
- Agentic AI layer with 8 callable tools and 4 distinct agentic behaviors
- Interactive Folium route maps per scenario per backend
- 6-tab Streamlit UI with KPI charts, scenario tables, and Excel export
- Pre-computed artifacts that auto-load on app startup
- Hugging Face Spaces deployment via Docker
- BackendConfig with 28 tunable parameters exposed in the sidebar
- Reactive event buttons for sick driver, urgent order, cancellation, and shift change

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Bento Overview

Layer	Description
Data Layer	40,000 orders · 400 routes · 8 depots · 4 goal profiles · auto-loaded from `./data/`
Scenario Layer	18 mutations across BASE, EAS_1-4, RT_1-7, GB_1-6 areas
Solver Layer	8 backends with consistent input/output contract via `_build_common_bundle`
Re-optimization Layer	Locked-prefix mechanism preserves stable assignments, only re-solves open work

Layer	Description
Agent Layer	Gemini 2.0 Flash with function-calling, autonomous tool chaining, max 10 iterations
UI Layer	Streamlit 6 tabs: Overview, KPI, Map, Scenario, Export, AI Agent
Artifact Layer	Per-backend Excel + Folium HTML + PNG previews saved to `./artifacts/`
Deployment Layer	Docker image deployable to HF Spaces or any container host

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The 8 Solver Backends — Benchmark Results

Tested on 150 orders · 20 routes · 18 scenarios from the included synthetic dataset.

Rank	Backend	Algorithm	Assigned	BASE km	Late min	OT min	Runtime s
🥇	**OR-Tools EA**	Split-solve CP + GLS	**150**	**231.8**	**0**	**0**	**770**
2	OR-Tools	CVRPTW + Guided Local Search	150	233.2	0	0	868
3	Esri	Tabu Search (cloud)	142	238.7	34	0	973
4	Adaptive	Multi-restart regret	150	424.1	0	0	6,472
5	Hybrid	Regret + LNS	150	424.1	0	0	1,790
6	Greedy	Nearest-neighbor + 2-opt	150	394.4	12,729	2,761	22
7	PyVRP	Hybrid Genetic Search	150	114.0	7,479	3,695	545
8	OSRM	Greedy + road distances	150	519.3	13,176	2,859	4,949

> **Important OR-Tools EA is the only backend that achieves all four production requirements:** 100% assignment, zero violations, sub-1000s runtime, and locked-prefix support.

The Innovation: Split-Solve Architecture

The OR-Tools Execution-Aware backend introduces a key architectural innovation. Locked orders **never enter the OR-Tools model** — they're copied verbatim from the baseline. Only open orders are submitted to the solver.

```
flowchart TD
    A[scenario_orders + scenario_routes + locked_prefixes] --> B

    B[Step 1: Separate locked from open<br/>locked_order_ids = baseline assignments<br/>open_orders_df = orders NOT in locked set]

    B --> C[Step 2: Anchor each vehicle<br/>last_locked.lat, last_locked.lon, last_locked.depart_time]

    C --> D[Step 3: Build OR-Tools model with ONLY open orders<br/>Capacity = MaxOrderCount minus n_locked<br/>PATH_CHEAPEST_ARC + GUIDED_LOCAL_SEARCH<br/>Full 45s budget]

    D --> E{Solution<br/>found?}

    E -->|yes| F[Step 4: Merge output<br/>locked stops verbatim from baseline<br/>open stops from OR-Tools<br/>Recompute timing from anchor]

    E -->|no or partial| G[Step 5: Greedy fallback<br/>Catches any dropped orders<br/>Guarantees 150 of 150 assignment]

    F --> G
    G --> H[Result: 150 of 150 assigned · 0 late · 0 overtime]

    style H fill:#EAF3DE,stroke:#3B6D11,color:#27500A
    style D fill:#E6F1FB,stroke:#185FA5,color:#0C447C
```

Why naive pinning fails

Submitting all 150 orders to OR-Tools with 130 of them having zero-slack time windows makes `PATH_CHEAPEST_ARC` infeasible. The solver fails to find any first solution within the time budget and returns 0 assignments.

By only submitting ~10 open orders, the model is **10-15x smaller**, the first solution comes instantly, and GLS gets the full 45s budget for improvement. Result: **11% faster than plain OR-Tools** and **0.6-3.4 km better** across all scenario groups.

Detection and Solver Stack

Solver Family Coverage

- **Constraint Programming:** OR-Tools, OR-Tools EA
- **Metaheuristics:** Hybrid, Adaptive, PyVRP
- **Greedy Heuristics:** Greedy + 2-opt, OSRM
- **Commercial Cloud:** Esri ArcGIS Network Analyst

Scenario Coverage

- **Execution-Aware (EAS_1-4):** Job timing changes mid-execution
- **Real-Time (RT_1-7):** Cancellations, sick drivers, urgent injects, shift delays
- **Goal-Based (GB_1-6):** Customer-specific weight overrides for travel/SLA/skill/overtime

Output Bundle (per backend)

- `scenario_summary` — 18 rows × 11 columns
- `route_output_all` — ~360 rows
- `stop_output_all` — ~2,700 rows
- `order_output_all`
- `scenario_actions`
- `scenario_timings`
- `requirement_checklist`
- `run_meta`

Agentic AI Layer

```
import google.generativeai as genai
from google.generativeai import protos
import os, json

class VRPOrchestrator:
    def __init__(self, inputs, config, existing_results=None, progress_cb=None):
        self.inputs = inputs
        self.config = config
        self.results = existing_results or {}
        self._tool_calls_log = []
        self._chat_session = None

        genai.configure(api_key=os.environ["GOOGLE_API_KEY"])
        self._genai = genai
        self._model = genai.GenerativeModel(
            model_name="gemini-2.0-flash",
            tools=self._build_gemini_tools(),
            system_instruction=self._build_system_prompt(),
        )

    def chat(self, user_message: str) -> str:
```

```
        if self._chat_session is None:
            self._chat_session = self._model.start_chat(
                enable_automatic_function_calling=False
            )
        return self._run_agent_loop(user_message)

    def _run_agent_loop(self, user_message: str) -> str:
        max_iterations = 10
        iteration = 0
        current_message = user_message

        while iteration < max_iterations:
            iteration += 1
            response = self._chat_session.send_message(current_message)

            fn_calls = []
            for candidate in response.candidates:
                for part in candidate.content.parts:
                    if hasattr(part, "function_call") and part.function_call.name:
                        fn_calls.append(part.function_call)

            if not fn_calls:
                # Pure text response — agent is done
                return " ".join(p.text for c in response.candidates
                                for p in c.content.parts if hasattr(p, "text"))

            # Execute all function calls
            fn_responses = []
            for fn_call in fn_calls:
                result = self._dispatch_tool(fn_call.name, dict(fn_call.args))
                self._tool_calls_log.append({
                    "tool": fn_call.name,
                    "input": dict(fn_call.args),
                    "result_preview": str(result)[:200],
                })
                fn_responses.append(
                    self._genai.protos.Part(
                        function_response=self._genai.protos.FunctionResponse(
                            name=fn_call.name,
                            response={"result": json.dumps(result, default=str)},
                        )
                    )
                )

            current_message = self._genai.protos.Content(
                role="user", parts=fn_responses,
            )
```

```
        return "Agent reached max iterations."
```

The 8 Agent Tools

Tool	Purpose
`get_scenario_list`	Returns all 18 scenario IDs and descriptions
`explain_scenario`	Explains what a scenario tests + what mutation it applies
`analyze_results`	Reads loaded bundles, surfaces metrics + violations
`detect_violations`	Finds late routes/orders + overtime amounts
`compare_backends`	Ranks backends by chosen priority criterion
`suggest_config`	Recommends BackendConfig overrides with reasoning
`run_backend`	Triggers a live solve and returns summary metrics
`trigger_reoptimization`	Re-runs with improved config, returns before/after diff

4 Agentic Behaviors

1. **Natural language interpretation** — `"Which backend is best for SLA?"`
2. **Autonomous re-optimization** — agent detects violations, picks config, re-runs, compares
3. **Multi-step reasoning** — `"Run A and B, then tell me which wins"` chains 3+ tool calls
4. **Real-time reactive** — one-click event buttons trigger autonomous handling

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Dependency Stack

```
streamlit>=1.28,<2.0
pandas>=2.0
numpy>=1.24
ortools>=9.7
folium>=0.15
streamlit-folium>=0.15
plotly>=5.18
openpyxl>=3.1
requests>=2.31
google-generativeai>=0.5
nbformat>=5.9
matplotlib>=3.7
python-dateutil>=2.8
```

Containerization Layer

```
FROM python:3.11-slim

RUN apt-get update && apt-get install -y --no-install-recommends \
    build-essential \
    && rm -rf /var/lib/apt/lists/*

RUN useradd -m -u 1000 user
USER user
ENV PATH="/home/user/.local/bin:$PATH"
WORKDIR /home/user/app

COPY --chown=user requirements.txt requirements.txt
RUN pip install --no-cache-dir --upgrade pip && \
    pip install --no-cache-dir -r requirements.txt

COPY --chown=user . .

EXPOSE 7860

CMD ["streamlit", "run", "app.py", \
     "--server.port=7860", \
     "--server.address=0.0.0.0", \
     "--server.headless=true", \
     "--server.enableCORS=false", \
     "--server.enableXsrfProtection=false", \
     "--browser.gatherUsageStats=false"]
```

Deployment Path

```
flowchart LR
    A[Local Dev + Benchmark Excel] --> B[Extract per-backend artifacts]
    B --> C[Pre-populate ./artifacts/ folder]
    C --> D[Streamlit app + auto-load logic]
    D --> E[Docker build]
    E --> F[Push to Hugging Face Spaces]
    F --> G[Public HTTPS endpoint]
    G --> H[Auto-loaded artifacts on first view]
    G --> I[Live solve via run_backend_suite]
    G --> J[Agent chat with GOOGLE_API_KEY secret]
```

Runtime Example

```
# Clone the repo
git clone https://github.com/dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio.git
cd Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio

# Create virtual environment
python -m venv vrp
source vrp/bin/activate

# Install dependencies
pip install -r requirements.txt

# Set free Gemini API key from aistudio.google.com (no credit card)
export GOOGLE_API_KEY="YOUR_GOOGLE_API_KEY"

# Run the app
streamlit run app.py --server.port 8504

# Open http://localhost:8504
# Auto-loads synthetic data + pre-computed artifacts
# Click "Build comparison" to see all 8 backends instantly
# Open the AI Agent tab to chat with Gemini
```

Hugging Face Deployment Pattern

- Build a Docker image with the Streamlit app and pre-computed artifacts
- Upload to Hugging Face Spaces (Docker SDK)
- Set `GOOGLE_API_KEY` as a Space secret
- Public HTTPS endpoint serves the auto-loaded studio
- Anyone can view benchmark results without running solvers
- Live re-solves available with one click per backend

```
git lfs install

git clone https://huggingface.co/spaces/AnubhaParashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio
cd Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio

# Copy the entire vrp_hf_package contents into this folder
git lfs track "data/orders_40000.csv"   # 7 MB file
git add .gitattributes
git add -A
git commit -m "Initial deploy"
git push
```

```
# Then in HF Settings → Variables and secrets:
#   GOOGLE_API_KEY = YOUR_GOOGLE_API_KEY
```

Public Demo Surfaces

> **Important** The Hugging Face Space provides instant browser-based access to the full studio with pre-loaded benchmark results. The GitHub repository contains the source code, architecture documentation, and reproducibility scripts.

Demo Links

- 🚀 **Live App:** AnubhaParashar / Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio
- 📦 **GitHub Repository:** Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio
- 📚 **Wiki Documentation:** Architecture · Algorithm Comparison · Technical Spec · Function Reference

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Demo Video

Embedded media: Open video

Algorithm Family Comparison

```
graph LR
    subgraph CP["Constraint Programming"]
        OR1[OR-Tools]
        OR2[OR-Tools EA · Winner]
    end

    subgraph META["Metaheuristics"]
        H[Hybrid]
        A[Adaptive]
        PV[PyVRP]
    end

    subgraph GR["Greedy Heuristics"]
        GD[Greedy + 2-opt]
        OS[OSRM]
    end

    subgraph CL["Commercial Cloud"]
        ES[Esri ArcGIS]
```

```
    end

    CP -.->|Hard TW · 100 percent assign · Zero violations| WIN[Production-grade]
    META -.->|Hard TW mostly · Slow · Custom rules| MIX[Specialized use]
    GR -.->|TW in cost only · Fast · Violations| BASE[Baseline only]
    CL -.->|Soft TW · Drops orders · Real roads| ENT[Enterprise GIS]

    style OR2 fill:#EAF3DE,stroke:#3B6D11,color:#27500A
```

Engineering Value

This project is not just about routing optimization.

It demonstrates how to take a combinatorial optimization problem and make it:

- reproducible (18 fixed scenarios + synthetic dataset)
- comparable (8 backends with identical I/O contract)
- execution-aware (locked-prefix preserves real-world state)
- agentic (Gemini autonomously detects and fixes violations)
- containerized (Docker image deployable to any host)
- publicly demoable (Hugging Face Space with pre-loaded artifacts)
- documented for reuse (Architecture, Algorithm Comparison, Function Reference)

Current Strengths

- Single Python codebase covers all 8 solver families uniformly
- Locked-prefix mechanism preserves in-progress assignments during re-optimization
- OR-Tools EA achieves 11% faster runtime + better distance than plain OR-Tools
- Agentic AI layer makes the system self-correcting without human dispatch input
- Pre-computed artifacts allow instant demo without running solvers
- Free Gemini API tier (1,500 req/day) keeps the AI layer zero-cost
- Reactive event buttons cover the 4 most common operational disruptions
- Excel export captures all 32 sheets for offline analysis
- Folium maps render route visualizations interactively per backend per scenario

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Next Improvements

- Add CI/CD pipeline for automated benchmark regression on new scenarios
- Persist OSRM cache to disk to survive Space restarts
- Add multi-depot solving support
- Add real-time technician GPS feed integration
- Add SLA breach prediction model alongside the routing solver

- Add A/B testing framework for cost function variants
- Add OpenAPI spec for headless solver invocation
- Move to Anthropic Claude as alternative agent backend
- Add Kubernetes deployment manifest for production scale-out
- Add load testing with realistic concurrent user patterns

Key Innovation

> **Important** This project connects **vehicle routing optimization** with **execution-aware re-planning** and **agentic AI orchestration** — three layers that are usually built separately.

It turns:

Static Route Planning → Execution-Aware Re-Optimization → Autonomous Violation Detection and Recovery

rather than stopping at one-shot solver output.

The OR-Tools Execution-Aware split-solve architecture is the keystone — it shows that **operational state (which stops are committed) is just as important as the optimization objective**.

Conclusion

This repository shows how a combinatorial optimization problem can evolve from a single solver invocation into a comparative benchmark studio with autonomous AI orchestration.

It combines:

- 8 solver backends with consistent I/O contract
- 18 reproducible operational scenarios
- Locked-prefix execution-awareness
- Google Gemini agentic AI with 8 callable tools
- Streamlit interactive UI with 6 specialized tabs
- Pre-computed artifacts for instant demo
- Docker deployment to Hugging Face Spaces
- Comprehensive technical documentation (Architecture, Algorithm Comparison, Function Reference)

Final Thought

> From **solving a routing problem** to **shipping an autonomous, execution-aware optimization platform**

> The real value is not only the OR-Tools EA winner backend — it is the **comparative benchmark scaffolding, the locked-prefix mechanism, and the agentic AI layer** that together make the system operationally trustworthy.

> 100% assignment · 0 violations · 770s runtime · 8 solvers compared · 18 scenarios stress-tested · 1 AI agent ready to handle the next disruption.